WHERE DO POWER SERIES WORK?
CLASSIFYING POINTS FOR SERIES SOLUTIONS

$$A(x)y'' + B(x)y' + C(x)y = 0$$

→ General form of 2nd order linear, homogeneous ODE.

By dividing the above ODE by $A(x)$, we get the ODE in standard form:

$$y'' + P(x)y' + Q(x)y = 0$$

where $P(x) = \frac{B(x)}{A(x)}$ & $Q(x) = \frac{C(x)}{A(x)}$

Importance of classifying points → Whether a point $x = x_0$ is "nice" or "problematic" depends only on how $P(x)$ & $Q(x)$ behave near $x = x_0$

Classifying points for series solutions of $y'' + P(x)y' + Q(x)y = 0$

ORDINARY POINT

Def^n $x = x_0$ is an ordinary point if both $P(x)$ & $Q(x)$ are analytic at x_0.

What it means: The equation is well-behaved at $x = x_0$; "nothing blows up".

What you do: Use an ordinary power series, i.e.,

$$y = \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

REGULAR SINGULAR POINT

Def^n $x = x_0$ is a singular point, where $(x - x_0)P(x)$ & $(x - x_0)^2 Q(x)$ are analytic at x_0.

What it means: $\frac{P(x)}{x - x_0}$ & $\frac{Q(x)}{(x - x_0)^2}$ "blow up" at x_0.

What you do: Use the Frobenius method.

$$y = (x - x_0)^r \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

where $a_0 \neq 0$, r is a non-integer or negative power of $(x - x_0)$

IRREGULAR SINGULAR POINT

Def^n $x = x_0$ is an irregular singular point, where $(x - x_0)P(x)$ & $(x - x_0)^2 Q(x)$ are not analytic at x_0

What it means: $\frac{P(x)}{x - x_0}$ & $\frac{Q(x)}{(x - x_0)^2}$ "blow up" at x_0.

What you do: Neither ordinary power series nor Frobenius method is guaranteed to work.

Quick summary:

Type of point	Test at $x = x_0$	Method
Ordinary "smooth road"	$P(x)$ & $Q(x)$ are analytic	Power series! $y = \sum_{n=0}^{\infty} a_n (x-x_0)^n$
Regular singular "pothole along the journey"	$P(x)$ & $Q(x)$ are not analytic, but $(x-x_0)P(x)$ & $(x-x_0)^2 Q(x)$ are analytic	Frobenius method! $y = (x-x_0)^r \sum_{n=0}^{\infty} a_n (x-x_0)^n$
Irregular singular "cliff, you can't get across"	$(x-x_0)P(x)$ & $(x-x_0)^2 Q(x)$ are not analytic	Neither method guarantees a solution

What "analytic" means → [Power series methods need both $P(x)$ & $Q(x)$ to be analytic!]

- "ANALYTIC" means the function, $f(x)$, [has a] convergent Taylor series at $x = x_0$.
- $f(x)\big|_{x=0} = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots$ → {$f(x)$ can be expressed as a power series for all x close to 0.}
- [$f(x)$ is smooth and well-behaved near $x = 0$] → There is no division by 0 or "blow up"

GENERAL FORM OF LINEAR, 2ND ORDER, HOMOGENEOUS ODE

$$A(x)\,y'' + B(x)\,y' + C(x)\,y = 0$$

LEGENDRE'S EQUATION

$$(1-x^2)\,y'' - 2xy' + n(n+1)\,y = 0$$

where $n = 1, 2, 3, \ldots$

Here:

$A(x) = 1-x^2$

$B(x) = -2x$

$C(x) = (n+1)n$

In standard form: $y'' + P(x)\,y' + Q(x)\,y = 0$

After dividing by $(1-x^2)$:

$$y'' - \left(\frac{2x}{1-x^2}\right)y' + \left(\frac{n(n+1)}{1-x^2}\right)y = 0$$

Here:

$P(x) = -\left(\frac{2x}{1-x^2}\right)$

$Q(x) = \frac{n(n+1)}{1-x^2}$

Check if $P(x)$ & $Q(x)$ are analytic at $x=0$:

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BESSEL'S EQUATION

$$x^2 y'' + xy' + (x^2 - \nu^2)\,y = 0$$

where $\nu = 0$ or positive real number

Here:

$A(x) = x^2$

$B(x) = x$

$C(x) = x^2 - \nu^2$

In standard form:

After dividing by x^2:

$$y'' + \left(\frac{1}{x}\right)y' + \left(1 - \frac{\nu^2}{x^2}\right)y = 0$$

Here:

$P(x) = \left(\frac{1}{x}\right)$

$Q(x) = \left(1 - \frac{\nu^2}{x^2}\right)$

Check if $P(x)$ & $Q(x)$ are analytic at $x=0$:

Analyticity of $P(x)$ & $Q(x)$ at $x=0$ in Legendre's equation

$$P(x) = -\left(\frac{2x}{1-x^2}\right) \qquad Q(x) = \frac{n(n+1)}{1-x^2}$$

$P(0) = 0$ $\qquad$ $Q(0) = n(n+1)$

→ $P(x)$ & $Q(x)$ are rational functions at $x=0$

→ $P(x)$ & $Q(x)$ can be written as power series at $x=0$

{ using a geometric series. $\frac{1}{1-u} = 1+u+u^2+\cdots$ for $0<u<1$ }

$$\Rightarrow P(x) = \frac{-2x}{1-x^2} = (-2x)\left[1+x^2+x^4+x^6+\cdots\right]$$

$$\Rightarrow Q(x) = \frac{n(n+1)}{1-x^2} = n(n+1)\left[1+x^2+x^4+x^6+\cdots\right]$$

Note: ① Nothing "blows up" at $x=0$

② $P(x)$ & $Q(x)$ are genuine power series near $x=0$

∴ $P(x)$ & $Q(x)$ are analytic & $x=0$ is an ORDINARY POINT

Where things break:

At $x=\pm 1$: $P(x)$ & $Q(x)$ "blow up"

∴ $x=\pm 1$ are singular points

∴ Series solutions are guaranteed $-1<x<1$

Analyticity of $P(x)$ & $Q(x)$ at $x=0$ in Bessel's equation

$$P(x) = \frac{1}{x} \qquad Q(x) = 1 - \frac{\nu^2}{x^2}$$

At $x=0$: $P(0)$ & $Q(0)$ "blow up"

∴ $P(x)$ & $Q(x)$ are not analytic at $x=0$

→ No power series can produce $P(x)=\frac{1}{x}$ & $Q(x)=\frac{1}{x^2}$ at $x=0$

Check analyticity of $xP(x)$ & $x^2Q(x)$:

$$xP(x) = x\,\frac{1}{x} = 1$$

$$x^2Q(x) = x^2\left(\frac{x^2-\nu^2}{x^2}\right) = x^2-\nu^2$$

Note: "$xP(x)$" & "$x^2Q(x)$" are polynomials:

→ Polynomials are already (finite) power series

∴ $xP(x)$ & $x^2Q(x)$ are analytic & $x=0$ is a REGULAR SINGULAR POINT

LEGENDRE'S EQUATION (continued)

$(1-x^2)\cdot y'' - 2x\cdot y' + n(n+1)\cdot y = 0$
where $n = 1, 2, 3, \ldots$

→ $x = 0$ is an ORDINARY POINT.

→ $P(x)$ & $Q(x)$ are analytic at $x = 0$

→ METHOD:
POWER SERIES
$y = \sum_{n=0}^{\infty} a_n x^n$

POWER SERIES METHOD
WORKS FOR ANY LINEAR, 2ND ORDER, ODE with ORDINARY POINT (x_0), AND ANALYTIC FUNCTIONS, $P(x)$, $Q(x)$

BESSEL'S EQUATION (continued)

$x^2 y'' + x y' + (x^2 - v^2) y = 0$
where $v = 0$ or positive real number

→ $x = 0$ is a REGULAR SINGULAR POINT.

→ $xP(x)$ & $x^2 Q(x)$ are analytic at $x = 0$

→ METHOD:
FROBENIUS METHOD
$y = x^r \sum_{m=0}^{\infty} a_m x^m$

r = real or complex number chosen so that $a_0 \neq 0$

FROBENIUS METHOD
WORKS FOR ANY 2ND ORDER LINEAR ODE, WITH REGULAR SINGULAR POINT (x_0), AND ANALYTIC FUNCTIONS $(x-x_0)P(x)$, $(x-x_0)^2 Q(x)$

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NATURE OF SINGULAR POINTS

EXAMPLE Determine the nature of the singular point of the following ODE:

$$x^2(1+x)y'' + x(4-x^2)y' + (2+3x)y = 0$$

Solution:

STEP-1 Write the ODE in standard form $\left[y'' + P(x)y' + Q(x)y = 0\right]$ & find the singular point.

Divide the ODE by $x^2(1+x)$. We have

$$y'' + \frac{x(4-x^2)}{x^2(1+x)}y' + \frac{2+3x}{x^2(1+x)}y = 0$$

$$\Rightarrow y'' + \frac{(4-x^2)}{x(1+x)}y' + \frac{2+3x}{x^2(1+x)}y = 0$$

$$\therefore \left[P(x) = \frac{4-x^2}{x(1+x)}\right] \left[Q(x) = \frac{2+3x}{x^2(1+x)}\right]$$

$\Rightarrow$ $\boxed{x=0}$ is a singular point

STEP 2: Determine the nature of the singular point:

Rewrite the standard form as $\left[y'' + \frac{p(x)}{x}y' + \frac{q(x)}{x^2}y = 0\right]$

$$\Rightarrow y'' + \frac{(4-x^2)/(1+x)}{x}y' + \frac{(2+3x)/(1+x)}{x^2}y = 0$$

$$\Rightarrow \left[p(x) = \frac{4-x^2}{1+x}\right] \& \left[q(x) = \frac{2+3x}{1+x}\right]$$

$\left[p(x) \& q(x) \text{ are analytic at } x=0 \text{ where } p(x) = xP(x) \& q(x) = x^2Q(x)\right] \Rightarrow \boxed{x=0}$ REGULAR SINGULAR POINT!

Further remarks:

$$\left[A(x)\,y'' + B(x)\,y' + C(x)\,y = 0\right]$$

$$\left[y'' + P(x)\,y' + Q(x)\,y = 0\right] \longrightarrow \text{After dividing by } A(x)$$

$$\left[y'' + \frac{p(x)}{x}\,y' + \frac{q(x)}{x^2}\,y = 0\right] \longleftrightarrow \text{where } p(x) = xP(x),\ q(x) = x^2 Q(x)$$

Note:

$p(x)$ & $q(x)$ are polynomials

where $p_0 = p(x)\big|_{x=0} = p(0) = \lim_{x\to 0} p(x) = \lim_{x\to 0} xP(x)$

$q_0 = q(x)\big|_{x=0} = q(0) = \lim_{x\to 0} q(x) = \lim_{x\to 0} x^2 Q(x)$